SE423 Software Project Management

Artifact 4: Stakeholder Analysis, Communications Plan, and Metrics

Scenario 11: National Employment and Skill Matching Platform

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Table of Contents

1. Stakeholder Register 3
2. Communication plan 5
3. Metric 9
3.1 Chosen Metrics 9
3.2 Metrics Explanation 9

1. Stakeholder Register

The Stakeholder Register identifies different individuals and groups that are involved or affected by the Fursah project. For each stakeholder it describes their relationships, influence, level of power and interests. This supports the team to know which stakeholder should be involved in the process, what information is needed, and how they should be managed throughout the project. Understanding the many stakeholders from the start helps the team be engaged with stakeholders, meet their expectations, and reduce issues that could affect the project's success.

Name & Title	Classification	Level of power	How stakeholders can affect the project	Level of interest (1-5)	Artifacts/metrics to communicate with stakeholder	Basic stakeholder management strategy to use
Fursah Team (Product Owner, Scrum Master, Developers, QA/Tester)	Internal	4	Develop, test, and deliver the platform. Their available resources and skills influence the quality and delivery of the program.	5	Product Backlog, Sprint plans, Lead time for changes, Change Failure Rate, Code Churn, Cyclomatic Complexity	Manage closely
Ministry of Human Resources and Social Development (HRSD)	Internal - Sponsor	5	Provides sponsorship and oversees the project, while reviewing and approving each release before it becomes accessible to the users.	5	Sprint review outcomes, release evaluations and approval	Manage closely
Job Seekers	External	2	Their use of the platform, from building profiles, finding job opportunities, monitoring applications progress these help determine the platform success.	5	New features showcased during Sprint Reviews and releases	Keep them informed

			Their Sprint Review feedback helps shape and guide the plan for the upcoming sprints.			
Students	External	1	Their use of the platform is to explore different career opportunities that match their skills.	4	New features and releases	Keep them informed
Employers - Large Firms and SMEs	External	3	Post jobs, oversee applications, evaluate candidates, schedule for interviews. Without the employers posting job postings They would have no job opportunities to offer for jobseekers.	5	New features showcased during Sprint Reviews and releases	Keep them informed
Universities and Training Providers	External	2	Provide and offer training and education programs that the platform suggest to users in Release 2.	3	Release 2 (Career Guidance and Training) plans	Monitor
Technology Firms	External	4	Provide the technical knowledge, tools, and implementation support in order to build and maintain the platform.	3	Technical requirements, release plans	Keep them satisfied
Other Government Agencies	External	4	Provide and supply the government services, authorization, and interfaces needed for the integration. Any delays from this affect Release 3.	3	Government integration plans (Release 3)	Keep them satisfied

2. Communication plan

Why the information is meaningful	Who it should be shared with	Communication format and tone	Who is responsible in the organization for communication	When and how frequently this communication will be made
1. Project Status and Progress: Communicating project status allows stakeholders to understand the overall progress of Fursah, completed activities, ongoing work, upcoming milestones, and any deviations from the planned schedule. It helps maintain transparency and ensures that project objectives remain aligned with stakeholder expectations.	HRSD, project team, and other stakeholders when the progress directly affects their involvement.	Formal progress reports shared through email, supported by in-person or virtual meetings when necessary.	Scrum Master, with progress updates provided by the development team.	At the end of every two-week sprint and whenever a significant milestone or delay occurs.
2. Sprint Results and Release Approvals: Sharing completed features allows stakeholders to evaluate the functionality delivered during each sprint, provide feedback, and verify whether the expected requirements have been met. HRSD also needs this information to review and approve releases before they become accessible to users.	HRSD, job seekers, students, and employers (large firms and SMEs), depending on the features demonstrated.	In-person or virtual Sprint Review demonstrations, accompanied by formal release evaluation reports and email summaries. The tone should be professional and formal.	Product Owner, supported by the Scrum Master and development team.	At the end of every two-week sprint during the Sprint Review. Formal release evaluations and approval requests are communicated before each release becomes accessible to users.
3. Risks and Issues: Communicating project risks and issues allows the team and relevant stakeholders to identify potential problems early, understand their possible impact on	HRSD, Product Owner, Scrum Master, and relevant development team members. Other	Formal email notifications for significant risks, supported by in-person or virtual meetings when	Scrum Master, with technical information and risk assessments	Risks are reviewed during each sprint. Critical issues are communicated immediately when identified, while ongoing

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project objectives, and take appropriate corrective or preventive actions. Critical risks must be escalated to avoid delays, security concerns, or disruptions to project delivery.	stakeholders are informed when a risk directly affects their involvement or interests.	discussion or decisions are required. The tone should be clear, formal, and objective.	provided by the relevant team members.	risks are reviewed during sprint meetings and planning activities.
4. Changes to Project Scope and Requirements: Communicating changes ensures that stakeholders understand modifications to project requirements, priorities, or deliverables. It prevents misunderstandings, maintains alignment with the original project objectives, and helps assess how proposed changes may affect the schedule and planned releases.	HRSD, Product Owner, Scrum Master, development team, and stakeholders directly affected by the proposed changes, including job seekers and employers when relevant.	Formal email notifications and in-person or virtual meetings for significant changes. Approved changes are documented in the Product Backlog. The tone should be formal and precise.	Product Owner, in coordination with the Scrum Master and development team.	Whenever a significant change is proposed or approved. Relevant updates are communicated during Product Backlog refinement and Sprint Planning.
5. Technical Quality and Project Metrics: Communicating technical quality measurements helps the team evaluate development performance, identify bottlenecks, monitor software maintainability, and detect deployment-related problems. The four selected metrics—	Development team, Scrum Master, and Product Owner. HRSD receives relevant quality and performance summaries when they	Technical metric reports, code review discussions, and Sprint Retrospective meetings. Relevant summaries are communicated through	Development team and Lead Developer for technical measurements, with the Scrum Master coordinating metric	Lead Time for Changes, Change Failure Rate, and Code Churn are reviewed every sprint. Cyclomatic Complexity is checked during code reviews and pull requests, with a consolidated

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Lead Time for Changes, Change Failure Rate, Code Churn, and Cyclomatic Complexity—provide measurable information that supports continuous improvement and helps maintain the quality of Fursah.	affect release evaluations or approval decisions.	email when necessary. The tone should be formal and technically precise.	reviews and communication.	summary at the end of each sprint.
6. Stakeholder Feedback and User Experience: Communicating stakeholder feedback helps the team understand user expectations, identify usability problems, and improve the platform's functionality. Feedback from job seekers, students, and employers is particularly important for improving the Job Seeker Portal, Employer Portal, and Smart Matching Engine.	HRSD, job seekers, students, and employers (large firms and SMEs), depending on the features being evaluated.	Feedback forms, interviews, usability testing sessions, and Sprint Review discussions. The tone should be professional, clear, and accessible to the intended audience.	Product Owner, supported by the Scrum Master and development team.	After relevant feature demonstrations and user testing sessions. Collected feedback is reviewed during Product Backlog refinement and considered during upcoming Sprint Planning sessions.
7. Release-Specific and Integration Updates: Communicating information about upcoming releases, technical requirements, and integration activities ensures that external stakeholders are prepared for the activities that involve them. This is particularly important for coordinating Career Guidance and Training activities and future government-service integrations.	Universities and Training Providers, Technology Firms, and Other Government Agencies, according to their involvement in the relevant release. HRSD is also informed about	Formal emails, technical documentation, and coordination meetings when technical clarification or joint decisions are required. The tone should be formal and professional.	Product Owner for release-related communication, supported by the Scrum Master and relevant technical team members for integration-related matters.	Before relevant release-planning activities and whenever integration requirements or timelines change. Updates are also provided when significant milestones are reached.

Why the information is meaningful	Who it should be shared with	Communication format and tone	Who is responsible in the organization for communication	When and how frequently this communication will be made
	release plans and significant integration milestones.			

The communications plan establishes how project-related information will be shared with stakeholders throughout the development of Fursah. Its purpose is to ensure that relevant stakeholders receive accurate and timely information according to their level of involvement and interest in the project. Since Fursah follows the Scrum methodology, communication will take place through regular sprint activities, formal updates, demonstrations, and stakeholder feedback sessions. The communication approach will be adapted to the nature of the information and the stakeholder receiving it.

Communication Approach

The project team will use Scrum activities as the primary means of coordinating internal work and communicating progress. Daily Scrums will support communication among team members, while Sprint Reviews will provide opportunities to demonstrate completed functionality and collect stakeholder feedback. Sprint Retrospectives will be used to discuss development performance, technical quality, and possible improvements to the team's working process.

Communication with external stakeholders will be tailored to their responsibilities and interests. HRSD will receive formal project progress updates and release evaluations because of its sponsorship and release approval responsibilities. Job seekers, students, and employers will receive relevant feature demonstrations, release information, and opportunities to provide feedback. Universities and Training Providers will be involved when Career Guidance and Training activities are planned, while Technology Firms and Other Government Agencies will receive technical and integration-related updates when their involvement becomes necessary.

3. Metric

3.1 Chosen Metrics

- Metric 1: Lead time for changes
- Metric 2: Change failure rate
- Metric 3: Code churn
- Metric 4: Cyclomatic complexity

3.2 Metrics Explanation

Metric	Rationale (why)	What measurements are to be taken	How often	By whom	How it is used	How team benef
1	Since working features are delivered every sprint (so that certain stakeholders can review).	Time it takes for a code change to go from commit to running in production.	Every sprint at the Sprint Retrospective.	Scrum Master and the lead developer	To find any bottlenecks, we will compare sprints. Any issue with waiting on testing or approval is included, and if a pattern is noticed, it will be discussed in the Retrospective.	We will be able to achieve the Definition of Done (DoD) for example through feedback from our stakeholders.
2	Due to Fursah's responsibility to ensure that personal and employment data are reliable. Showing how frequent problems introduced with releases are.	Percentage of deployments that cause a failure in production.	Every sprint (at the end of each release, we have a cumulative view).	QA/Tester and lead developer	High rate= weak DoD, meaning that the team should tighten code reviews or even add regression tests. Product Owners are then shown these results when we do the Sprint Review.	We could benefit from having less incidents in production, and an increase in stakeholder trust.
3	In every sprint, we will probably receive feedback that ends in changes in our	Over a time window, it's the sum of the added lines, modified lines, and deleted LOC.	Every sprint	Scrum Master (reviews) and developers	Unclear requirements and unstable design is indicated when we get high churn.	We would be able to detect any instability early on, as well as not wasting efforts

	requirements. Churn entails the instability and rework of our code.					on repeatedly rewritten code.
4	To keep our code testable and maintainable, especially the more complex parts.	A product metric that measures the cyclomatic complexity for each function. We usually aim for less than or equal to 10 per function, and it will be refactored if its more than 15	Every time we do a code review or pull request (with a summary every sprint)	Developers and the reviewing developer checks it	High complexity code should be simplified (or given more test cases). We use it as a quality gate when we review our code	It will help us by allowing our code modules to have less defects, and overall making easier and testing and maintenance.